

**CONESTOGA COLLEGE**  
**INSTITUTE OF TECHNOLOGY and ADVANCED LEARNING**  
Electronics Engineering Technology (Telecommunications) & Computer Engineering Technology

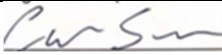
**CAPSTONE PROJECT I**

**PROJECT CONCEPT PROPOSAL & REQUIREMENTS**

**Essential's Emporium**



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|-----------------|----------------------------------------------------------------|--|--|-----------------|
| Document Title  | Concept Proposal & Requirements Template<br><i>Signal Sync</i> |  |  |                 |
| Document Number | CP-C1                                                          |  |  | Revision<br>A   |
| Date Issued     | 2025-04-10                                                     |  |  | Replaces<br>NEW |

|         |                 |                   |                                                                                     |            |
|---------|-----------------|-------------------|-------------------------------------------------------------------------------------|------------|
|         | Name            | Title/Expert Role | Signature                                                                           | Date       |
| Authors | Edward Estacion | Software Designer |  | 2024-12-05 |
|         | Carson Soers    | Hardware Designer |  | 2024-12-05 |

REVISION HISTORY

|          |                                                        |                |
|----------|--------------------------------------------------------|----------------|
| Revision | Description of Change                                  | Effective Date |
| NEW      | The initial release of Concept & Requirements Proposal | 2024-12-05     |
| REV A    | The demonstration update                               | 2025-04-10     |
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# Signal Sync

## 1.0 PURPOSE

The purpose of this document is to outline the concept and requirements for the Signal Sync electronic tracking device. This device will provide a simple way to locate other individuals at large public events. By collecting one of our devices and inputting a setting based on your interest, you open a fun and easy way to locate others who have selected the same setting.

## 2.0 SCOPE

This document is intended to satisfy in part, the concept and requirements for the Capstone Project for Terms 5 and 6 and Co-Op Terms 9 and 10.

## 3.0 NEED STATEMENT

The current landscape for connecting with others at an event is difficult. We propose a new take on forming connections that removes the need for a smartphone and an internet connection. Simply collect one of our devices and select your interest to begin broadcasting your location and receiving locations of others with the same interest.

## 4.0 GOAL

Our vision is to provide a fun and unique way to strike up a conversation with others in a large event setting. Our device removes the barrier of insecurity when approaching a stranger by letting others know you want to chat, and so do they.

## 5.0 OBJECTIVES

The objective(s) of this project is(are) to:

- 5.1 Create a device capable of transmitting and receiving an approximate location.
- 5.2 Only communicate with like devices that have input the same setting.
- 5.3 Output the approximate location to a screen to indicate others' positions relative to you.
- 5.4 Use sub-1GHz frequency for communication to avoid saturated bands.
- 5.5 Use on-board alerts to notify users of location pings

## 6.0 DEFINITIONS

**Raspberry Pi Zero W:**

- A small, low-power, single-board computer with built-in Wi-Fi and Bluetooth, used as the main processing unit in your project.

**EFR32G28**

- Radio Board by Silicon Labs

**UART**

- A communication protocol for serial data transmission, commonly used to send and receive data between microcontrollers and computers.

**Mesh Network:**

- A network topology where devices (nodes) connect directly to one another, relaying data between nodes to increase range and reliability.

**RF (Radio Frequency):**

- The use of electromagnetic waves to transmit data wirelessly. RF communication is used for location tracking and broadcasting in this project.

**RSSI (Received Signal Strength Indicator):**

- A measurement of the power of a received RF signal, often used to estimate the distance between devices in RF-based location tracking.

## **7.0 INTRODUCTION**

This project aims to fill the need for a way to find others with the same interest as you at a large event with many attendees. In the modern age of technology and electronics people can feel lost in all the options available for meeting others. These methods often have an awkward connotation attached to them and don't offer a more simplistic approach to talking with strangers.

We propose to remove the stigma of using apps and online connections to one that offers a fun device that indicates what you're interested in and that you want to talk about it. There is no need for an initial awkward meet-up that involves looking for a specific person in a large crowd but simply tracking down the individual with a Signal Sync device set to the same interest, clearly shown by an LED indicator.

The competition for our product is largely cornered by the online space and using apps and online forums to try to make a connection. Our device removes the need to reach out online, where you're blindly putting yourself out there. We offer a way to meet up with others that are actually attending the event and it's a fun way to find them.

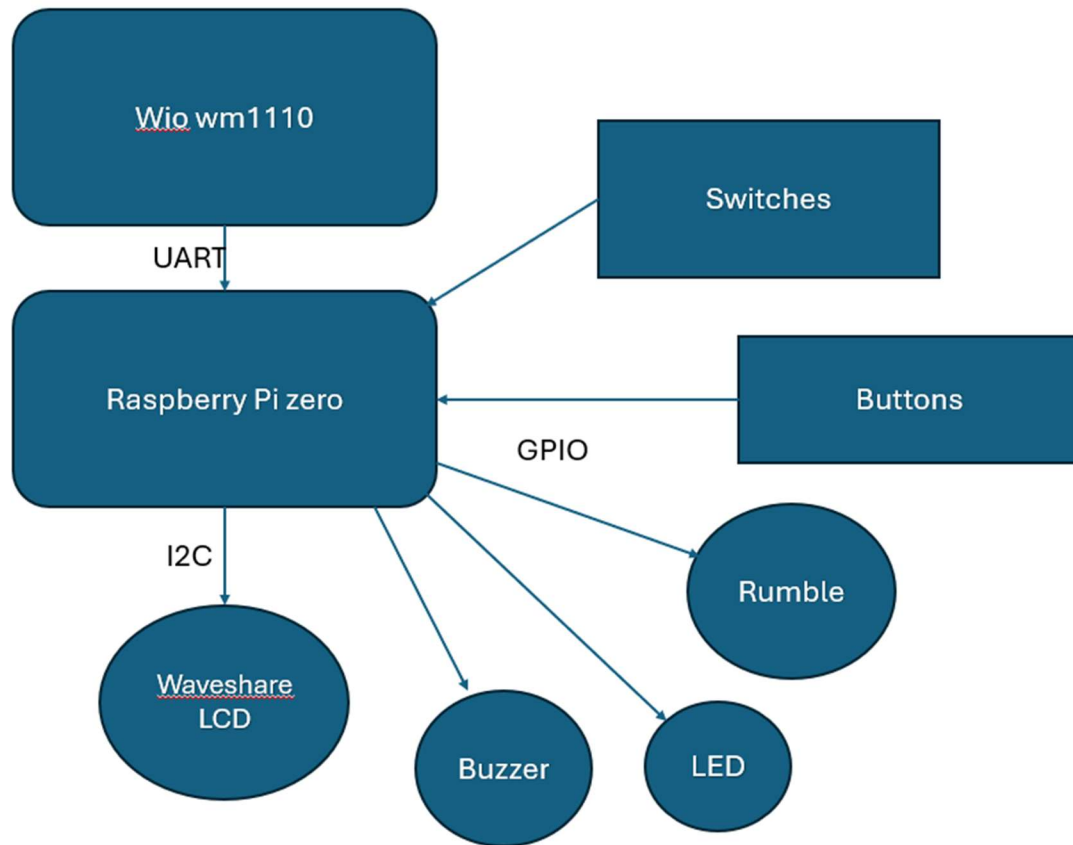
## **8.0 PROJECT ORGANIZATION**

## 8.1 Project Overview

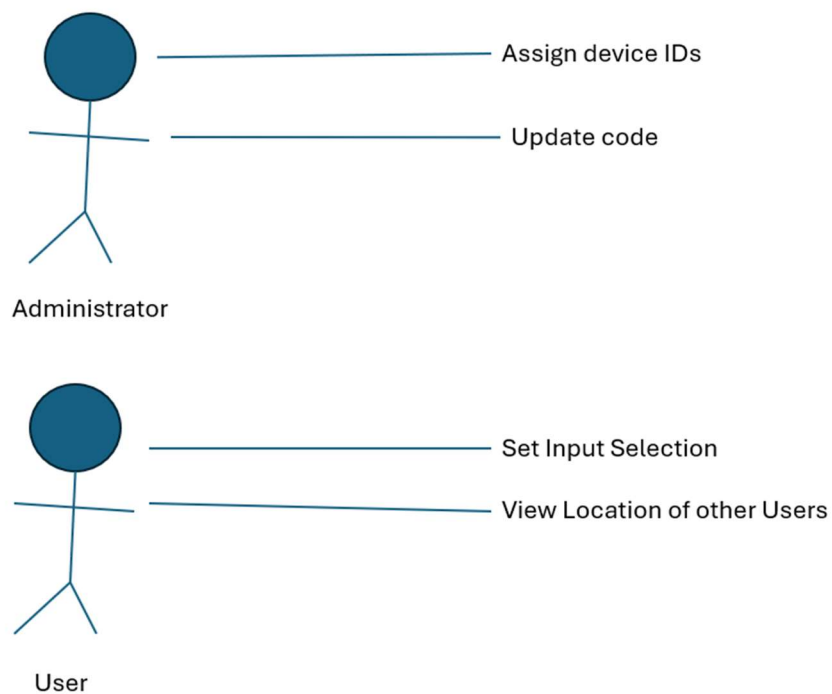
Concept Image



## 8.2 High-Level Flow Chart

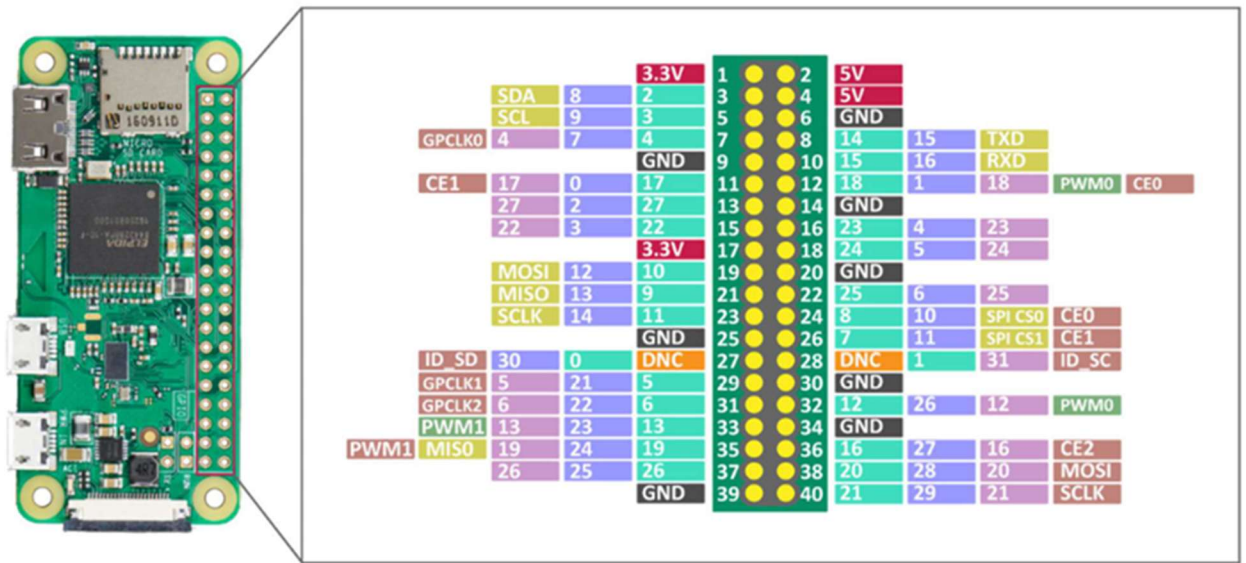


### 8.3 UML Use Diagram

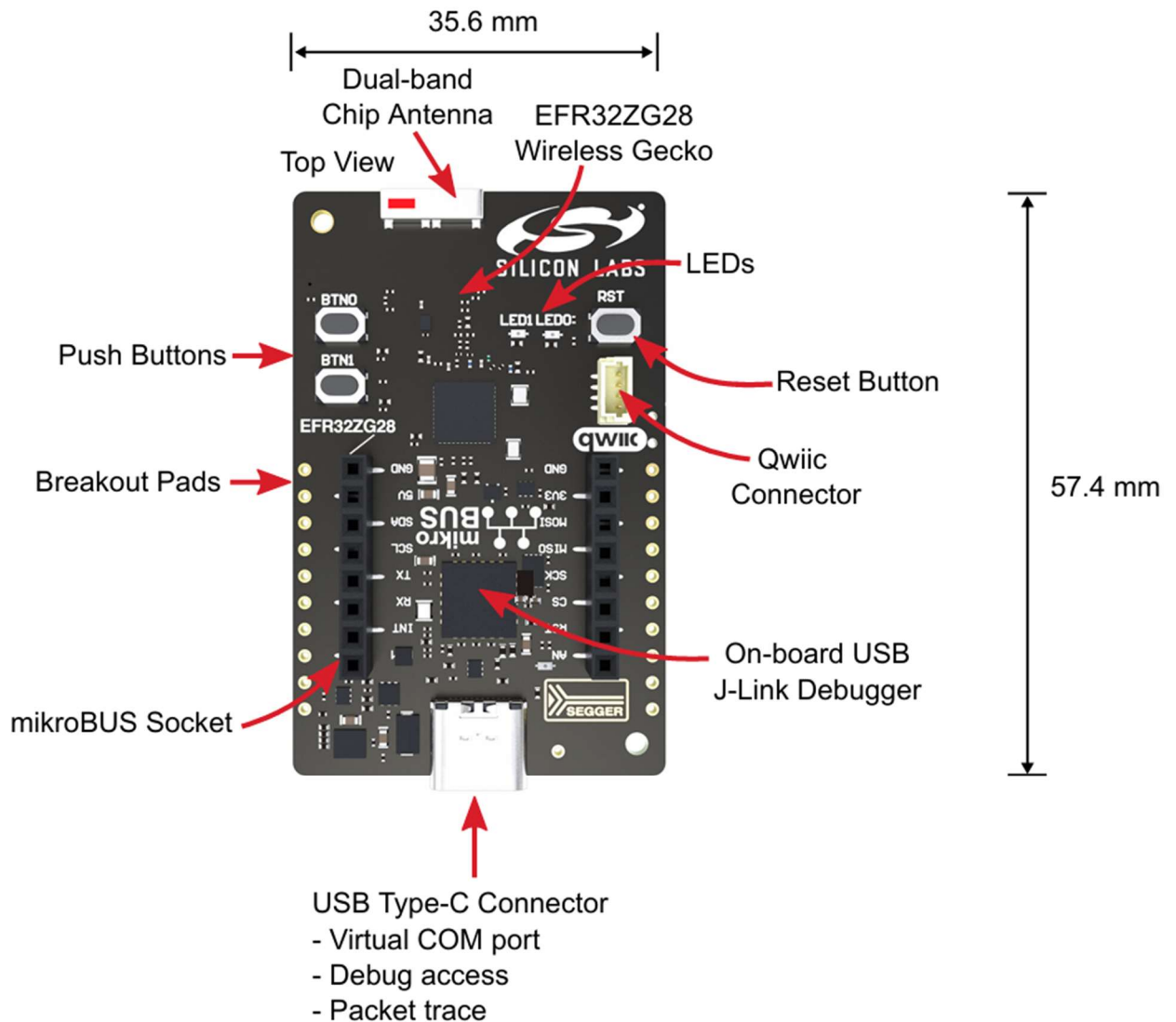


### 8.4 Hardware and Software Modules

## Pi Zero W



## EFR32ZG28 EVALUATION BOARD





## 9.0 PRODUCT USE, USERS and USERS NEEDS

This product is intended for the following purposes, activities, or applications:

### 9.1 Intended Use

- 9.1.1. To broadcast and receive a sub 1GHz signal revealing location
- 9.1.2. To be able to a signal unique to the user and the interest that is selected

### 9.2 Intended Users

The product is intended to be used by the following personnel:

- 9.2.1. Anyone who is attending a public event and is interested in meeting others

### 9.3 User Needs

- N1. Input a setting that represents an interest in a topic or activity
- N2. Ability to track other devices with the same setting input
- N3. Visual representation of the tracking to a screen
- N4. Visual indication of the other devices with physical LEDs
- N5. Audible and physical notifications of others like devices detected
- N6. Ability to mute another like device set to the same interest

## 10.0 REQUIREMENTS

### 10.1 Functional Requirements

- Q1. Provide real-time tracking of other like devices
- Q2. Be part of a network in which all like devices can detect one another
- Q3. Operate on a sub-1GHz frequency
- Q4. Mute a device that you no longer wish to get location updates from

### 10.2 Performance Requirements

- Q5. The device behaves like a mesh network with other similar devices
- Q6. The device can work indoors
- Q7. The device has a sufficient range to accommodate the footprint large events
- Q8. The device can support many other devices in this same network

### 10.3 Safety Requirements

- Q9. Protection Against Electrical Shock: Use proper grounding and ensure all components are properly insulated to prevent electrical shocks.
- Q10. Battery Safety: Use of certified rechargeable batteries, overcurrent and overheating protection.
- Q11. Heat Management: Ensure heat sinks are placed to avoid overheating hardware components.

### 10.4 Standards Compliance Requirements

- Q12. IEC 62368-1:
  - A global standard for the safety of electronic devices, including low-power systems like yours.
- Q13. UL 60950-1 (Deprecated, replaced by IEC 62368-1):



- Applies to IT and communications equipment, ensuring safe operation.  
Q14. **IEC 62133:**
- Covers safety for rechargeable batteries, ensuring safe operation during charging and discharging.  
Q15. **ISO 19115:**
- Covers metadata for geographic information and tracking data.  
Q16. **RSS-Gen:**
- Provides general requirements for RF devices, including fundamental operating conditions, labeling, and emission limits.

## **11.0 CONSTRAINTS (Critical)**

C1. The device will transmit and receive an RSSI signal at a sub-1GHz frequency

C2. The device will operate in a mesh with other similar devices

C3. The device will display an approximate location of other similar devices that are set to the same input based on an interest

## **12.0 PROJECT SPONSOR**

### **12.1 Contact Person and Contact Information**

Andrew Marchment

[Amarchment@telesat.com](mailto:Amarchment@telesat.com)

226 929-2433

## **13.0 CAPSTONE PROJECT LEAD PERSON and CONTACT INFORMATION**

### **13.1 Contact Name**

Carson Soers

### **13.2 Contact Information (address, Tel. No., e-mail, etc.)**

Carson Soers, 519 901 2557, [csoers1106@conestogac.on.ca](mailto:csoers1106@conestogac.on.ca)

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## **14.0 DESIGN TEAM**

### **14.1 Team Members and Roles**

Carson Soers – Project Lead – Hardware Specialist

Edward Estacion – Software specialist

### **14.2 Team Rules of Operation**

Commitments:

As a project team we will:

1. Maintain honesty and realism in project planning.
2. Only undertake in work that the team can do.

3. Ensure confidentiality of proprietary information.
4. Operate proactively to anticipate and prevent potential problems.
5. Operate within the expectation that work will be split evenly.
6. Focus on what is best for the project.
7. See that the project is completed within the time constraints.

#### Team Meetings:

##### Participation We will:

1. Give everyone an equal opportunity to participate.
2. Encourage each other to bring our creative thoughts on all topics.
3. Be open to listening to new approaches and ideas.
4. Be honest with the other members of the team.
5. Avoid placing blame, us against the problem not against each other.

#### Team Meetings:

##### Communication We will:

1. Keep focused on the project.
2. Be as clear as possible.
3. Do your best to understand.
4. Respect others' ideas and see topics from their perspective.
5. Use visual aides to assist in creating discussion.

#### Team Meetings:

##### General Process We will:

1. Address problems as they arise and build on each others' ideas of solutions.
2. Encourage all ideas, respecting that not every problem has a cookie cutter solution.
3. Represent problems and solutions as data points to be as clear as possible.
4. Use a creative process to expand our horizons.
5. When a decision is made, ensure that it makes sense.
6. Use data to assist us in making decisions.
7. Discuss the impact of the decision before making it.
8. Get input from the entire team before a decision is made.
9. Ask all team members if they support a decision before it is made.
10. Treat conflicts as an opportunity for understanding.
11. Choose a time and place to discuss major issues.
12. Ensure we understand all the points being made.
13. Seek to find the best solution through compromise and creativity.

#### Team Meetings:

##### Meeting Guidelines

1. Meetings will be held weekly.
2. The meeting will be called by the team leader.
3. Meeting agendas will be issued 2 days in advance by a collaboration of members.
4. Meetings will be either online or in person.

#### Team Meetings:

##### Meeting Procedures

1. Meetings will begin on time.
2. Team members will come to the meetings prepared.
3. The meeting agenda will be followed.
4. A list will be created during the meeting to plan future work.

5. Meetings tasks will be rotated among members.

## 15.0 References

- [1] IEEE. (V9 2016-07-22). IEEE Editorial Style Manual [Online]. Available: [https://www.ieee.org/publications\\_standards/publications/journalmag/IEEE\\_style\\_manual.Pdf](https://www.ieee.org/publications_standards/publications/journalmag/IEEE_style_manual.Pdf)
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